SAT Math Mastery · Session 8

Data Models, Angles and Triangles

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 45 minutes), and a short error log.
Topics	scatterplots and lines of best fit: predicted versus actual, slope and intercept in context, linear versus exponential models · parallel lines and transversals · angle sums and exterior angles · isosceles triangles and regular polygons · similar triangles and proportional segments · triangle inequality, midsegments and angle bisectors · right-triangle trigonometry and special right triangles · Law of Cosines
Name / Date	_____ _____

PART A · IN CLASS — 37 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. Medium · Desmos

A household water-conservation study recorded, for several summer days, the daily high temperature x (in degrees Celsius above 25 °C) and the household's water use y (in liters). A line of best fit for the data is given by $y = 12x+20$. On one recorded day, $x = 10$ and the household's actual water use was 155 liters. What is the positive difference between the actual water use on that day and the water use predicted by the line of best fit?

2. Medium · Desmos

A scatterplot shows the relationship between x, the distance in kilometers from a city's central waterfront, and y, the number of visitors counted per hour at a beach access point. The line of best fit for the data passes through the points $(2, 340)$ and $(10, 60)$. What is the slope of this line of best fit?

(A) 35

(B) -35

(C) -28

(D) 28

3. Medium · Desmos

A line of best fit models y, the installed solar capacity (in megawatts) at a pilot site, in terms of x, the number of years since 2015: $y = 45.2x + 12.5$. Which of the following is closest to the capacity predicted by this model at $x = 8.4$?

(A) 380.5

(B) 388.7

(C) 392.2

(D) 401.0

4. Medium · No calculator needed

A water-treatment plant's storage tank begins a shift completely full. The tank is drained at a constant rate throughout the shift to supply a neighborhood. What type of model best describes the volume of water remaining in the tank at any time during the shift?

(A) A linear model with a negative slope

(B) An exponential growth model

(C) An exponential decay model

(D) A linear model with a positive slope

5. Medium · No calculator needed

A scatterplot shows exam-preparation hours x and practice-exam score y for 10 students, with a line of best fit shown. For the student with the highest practice-exam score, the actual score was 96, and the score predicted by the line of best fit was 89. Which of the following is closest to the difference between the actual score and the predicted score for this student?

(A) 4

(B) 1

(C) 10

(D) 7

6. Medium · Desmos

The table shows the age x, in years, and the efficiency loss y, in percent, for a sample of rooftop solar panels.

x	1	2	3	4	5
y	3.5	6.0	8.6	11.0	13.4

Which of the following equations could be the line of best fit for the data?

(A) $y = -2.5x + 1$

(B) $y = 2.5x - 1$

(C) $y = 2.5x + 1$

(D) $y = x + 2.5$

7. Hard · Desmos

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

(A)

x	1	2	3	4
y	5	10	15	20

(B)

x	1	2	3	4
y	3	6	12	24

(C)

x	1	2	3	4
y	5	9	13	17

(D)

x	1	2	3	4
y	2	4	6	8

8. HARD · NO CALCULATOR NEEDED

A line of best fit models the relationship between x, the number of years since 2018, and y, the average monthly electricity subsidy, in dollars, provided to households in an energy-efficiency pilot program: $y = -18.6x + 340$. Which of the following is the best interpretation of the slope of this line of best fit in this context?

(A) The estimated subsidy decreases by \$340 each year.

(B) The estimated initial subsidy was \$18.6.

(C) The estimated subsidy increases by \$18.6 each year.

(D) The estimated subsidy decreases by \$18.6 each year.

9. HARD · DESMOS

Sam deposits \$800 into Account A, which earns 5% interest compounded annually. Sam also deposits \$1,500 into Account B, whose balance increases by a fixed \$40 each year. Based on these growth patterns, which of the following best describes the long-term behavior of the balances of Account A and Account B?

(A) The balance of Account A starts lower than the balance of Account B but eventually exceeds it.

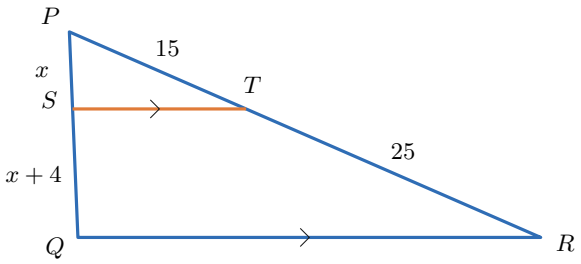
(B) The balance of Account A starts higher than the balance of Account B but eventually falls below it.

(C) The balance of Account A is always greater than the balance of Account B.

(D) The balance of Account A is always less than the balance of Account B.

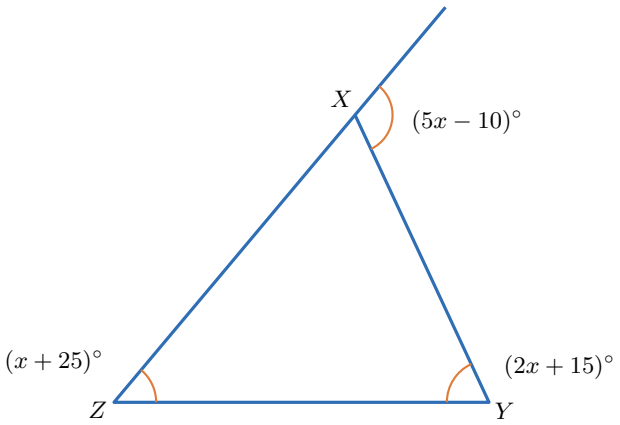
10. HARD · DESMOS

In triangle PQR, segment ST is drawn parallel to side QR, where S lies on PQ and T lies on PR. If $PS = x$, $SQ = x + 4$, $PT = 15$, and $TR = 25$, what is the value of x?



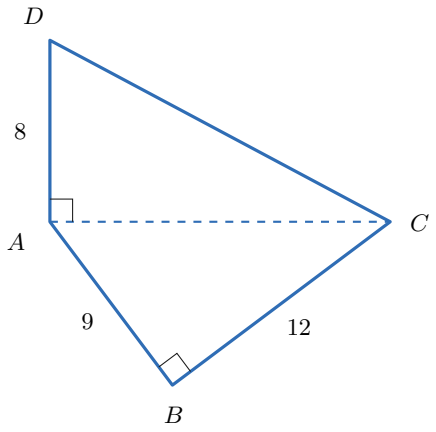
11. HARD · DESMOS

In triangle XYZ, the exterior angle at vertex X measures $(5x - 10)°$. This exterior angle equals the sum of the two non-adjacent (remote) interior angles, which measure $(2x + 15)°$ at vertex Y and $(x + 25)°$ at vertex Z. What is the measure of angle Y, in degrees?



12. HARD · DESMOS

In the figure, quadrilateral $ABCD$ is formed by two right triangles, ABC and ACD, sharing diagonal AC. Triangle ABC has a right angle at B with $AB = 9$ and $BC = 12$. Triangle ACD has a right angle at A with $AD = 8$. What is the area of quadrilateral $ABCD$?



(A) 108

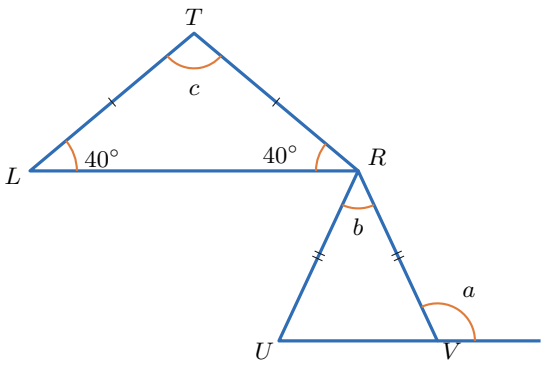
(B) 114

(C) 90

(D) 96

13. Hard · Desmos

In the figure shown, two isosceles triangles share the vertex R. In the top triangle, the vertex angle is c and each of its two equal base angles measures $40°$. In the bottom triangle, the vertex angle b is at R and equals half of c. Angle a is formed by extending the base of the bottom triangle, so a is supplementary to one of the bottom triangle's two equal base angles. What is the value of a, in degrees?



14. Hard · Desmos

In triangle ABC, the exterior angle at vertex C measures $130°$. If the measure of angle A is $55°$, what is the measure of angle B?

15. Hard · Desmos

Lines l and m are parallel and are cut by transversal t. Which of the following pairs of measures could NOT represent the measures of two same-side interior angles formed by l, m, and t?

(A) $95°$ and $85°$

(B) $112°$ and $68°$

(C) $95°$ and $84°$

(D) $100°$ and $80°$

16. Hard · Desmos

In triangle ABC, the exterior angle at vertex C measures $(7x - 12)°$. This exterior angle equals the sum of the two non-adjacent interior angles, which measure $(3x + 8)°$ and $(2x + 16)°$. What is the value of x?

17. Hard · Desmos

A pole that is 2.4 meters tall casts a shadow 3.2 meters long. At the same time, a nearby communications tower casts a shadow 52 meters long. Using similar triangles, which of the following is closest to the height, in meters, of the tower?

(A) 39.0

(B) 41.6

(C) 34.7

(D) 36.4

18. Hard · Desmos

In isosceles triangle ABC, $AB = AC$, and the measure of angle A is $30°$ more than the measure of angle B. What is the measure, in degrees, of angle C?

19. Hard · Desmos

In triangle ABC, points M and N are the midpoints of sides AB and AC, respectively. If $MN = 3x - 4$ and $BC = 5x + 2$, what is the length of BC?

(A) 56

(B) 42

(C) 47

(D) 52

20. Hard · Desmos

In $\triangle XYZ$, the side lengths are a, b, and c, where $4 < a < 6$ and $9 < b < 11$. Which of the following represents all possible values of c?

(A) $5 < c < 17$

(B) $3 < c < 15$

(C) $3 < c < 17$

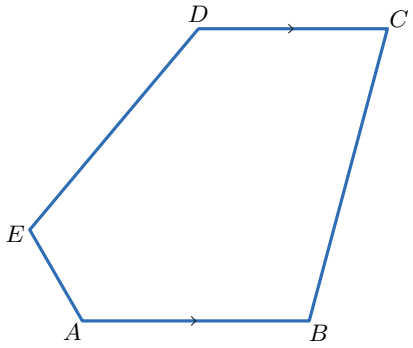
(D) $5 < c < 15$

21. HARD · DESMOS

In triangle ABC, sides AC and AB are congruent. Points D, C, and B are collinear, with C between D and B. If the measure of angle DCA is $112°$, what is the measure, in degrees, of angle A?

22. HARD · DESMOS

In convex pentagon $ABCDE$ shown (not drawn to scale), side $\overline{AB}$ is parallel to side $\overline{DC}$. The measures of four interior angles are $\angle A = (6x)°$, $\angle B = (5x + 5)°$, $\angle C = (3x + 15)°$ and $\angle D = (7x - 10)°$. What is the measure, in degrees, of $\angle E$?



(A) 130

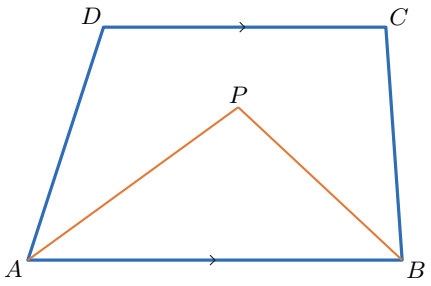
(B) 110

(C) 70

(D) 105

23. HARD · DESMOS

In trapezoid $ABCD$ shown (not drawn to scale), $\overline{AB} \parallel \overline{DC}$, $\angle D = 108°$, and $\angle C = 94°$. The bisector of $\angle DAB$ and the bisector of $\angle ABC$ intersect at point P inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at P by these two bisectors?



(A) 79

(B) 22

(C) 108

(D) 101

24. HARD · DESMOS

Two lines intersect at point O, and the smallest of the four angles they form measures $50°$. A ray from O bisects each of the four angles. Together with the two lines, this gives eight rays from O, which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

(A) $140°$

(B) $115°$

(C) $130°$

(D) $155°$

25. HARD · DESMOS

In equilateral triangle ABC, point D lies on side $\overline{BC}$ such that $BD = \frac{1}{4}BC$. What is the value of $\tan\angle DAC$?

(A) $\frac{3\sqrt{3}}{7}$

(B) $\frac{3\sqrt{3}}{5}$

(C) $\frac{\sqrt{3}}{7}$

(D) $\frac{5\sqrt{3}}{9}$

26. HARD · DESMOS

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. The measure of $\angle ADE$ is $(4x - 10)°$, the measure of $\angle ABC$ is $(2x + 30)°$, and the measure of $\angle ACB$ is $(x + 25)°$. What is the measure, in degrees, of $\angle BAC$?

(A) 115

(B) 45

(C) 65

(D) 70

27. HARD · DESMOS

Points A, B, C and D are consecutive vertices of a regular polygon with n sides, and $\overline{AC}$ is a diagonal. The measure of $\angle BAC$ is $(x + 3)°$ and the measure of $\angle ACD$ is $(10x + 15)°$, where x is a constant. What is the value of n?

(A) 12

(B) 15

(C) 24

(D) 8

28. Hard · Desmos

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle $120°$ and common difference $5°$. How many sides does the polygon have?

(A) 16

(B) 18

(C) 7

(D) 9

29. Hard · Desmos

Two lines intersect and form four angles. The acute angles each measure $(2k + 4)°$, where k is a constant. Each obtuse angle measures 4 degrees more than 3 times the measure of an acute angle. Which of the following CANNOT be the sum, in degrees, of the measures of two of the four angles?

(A) 88

(B) 180

(C) 224

(D) 272

30. Hard · Desmos

In triangle ABC, the measure of angle A is $(3x + 11)°$, the measure of angle B is $(2x + 4)°$, and the measure of angle C is $30°$. What is the value of

$$(\sin A \cos B + \cos A \sin B)(\cos A \cos B - \sin A \sin B)\,?$$

(A) $\dfrac{1}{4}$

(B) $-\dfrac{\sqrt{3}}{4}$

(C) $\dfrac{\sqrt{3}}{4}$

(D) $-\dfrac{1}{4}$

31. HARD · DESMOS

In triangle ABC, $AB = 12$ and $AC = 18$. The bisector of angle A meets $\overline{BC}$ at D, and $BC = 20$. What is the length of the longer of the two segments BD and DC?

32. HARD · DESMOS

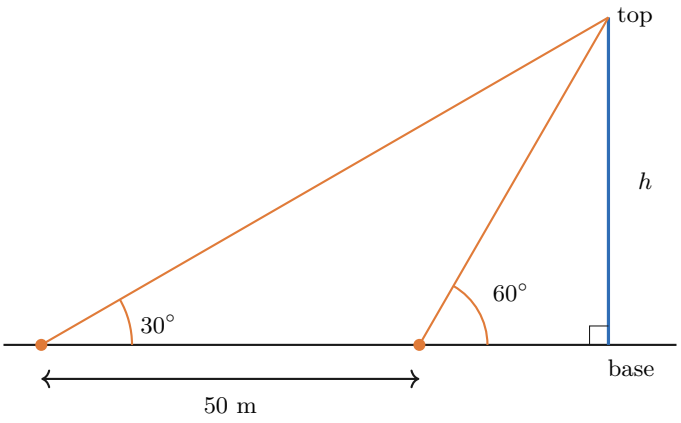
Two sides of a triangle have lengths 7 and 12. The third side has an integer length. How many different integer lengths are possible for the third side?

33. HARD · DESMOS

In triangle ABC, $AB = AC$ and angle A measures $40°$. Point D lies on $\overline{AC}$ such that $BD = BC$. What is the measure, in degrees, of angle ABD?

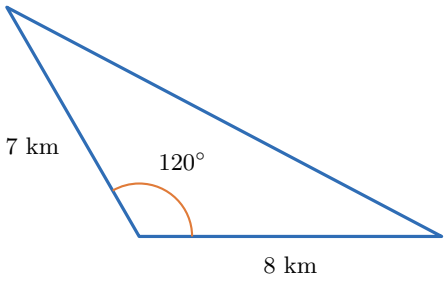
34. HARD · DESMOS

Engineers surveying a proposed tower measure the angle of elevation to the top of the tower from a point on the ground to be 30°. Standing 50 meters closer to the base of the tower, along the same straight line, the angle of elevation is 60°. What is the height of the tower, in meters, to the nearest tenth?



35. HARD · DESMOS

A triangular plot of land has two sides of length 7 kilometers and 8 kilometers, with an included angle of 120° between them. What is the length, in kilometers, of the third side of the plot?



(A) 15

(B) 11

(C) 12

(D) 13

36. HARD · DESMOS

In right triangle JKL, the right angle is at L. If $\cos(J) = \frac{7}{25}$, what is the value of $\sin(J)$?



(A) $\frac{24}{25}$

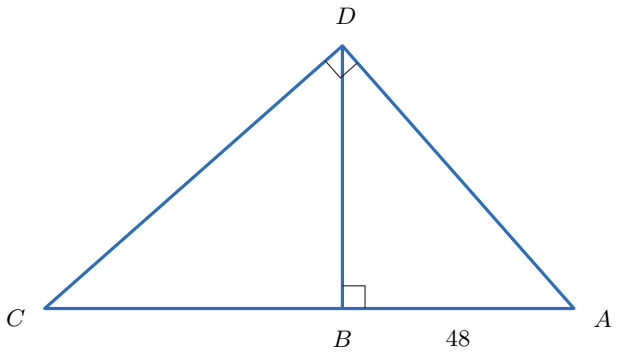
(B) $\frac{25}{7}$

(C) $\frac{25}{24}$

(D) $\frac{7}{25}$

37. HARD · DESMOS

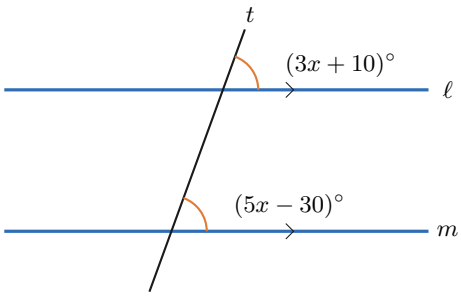
In the figure shown, right triangle ACD has a right angle at D, and right triangle BCD, with B on segment AC, has a right angle at B. Both triangles share angle C. If $\cos C = 0.75$ and $AC = 48$, what is the length of BC?



PART B · SELF-ASSESSMENT — 24 PROBLEMS

38. MEDIUM · DESMOS

Line ℓ is parallel to line m and both are intersected by transversal t. One angle formed at the intersection with ℓ measures $(3x+10)°$, and its corresponding angle at the intersection with m measures $(5x-30)°$. What is the value of x?



(A) 15

(B) 20

(C) 40

(D) 10

39. MEDIUM · DESMOS

In triangle ABC, the measure of angle A is twice the measure of angle B, and the measure of angle C is $24°$ more than the measure of angle B. What is the measure of angle A, in degrees?

(A) 39

(B) 52

(C) 78

(D) 87

40. Medium · Desmos

A regular polygon has exterior angles that each measure 24°. How many sides does the polygon have?

(A) 15

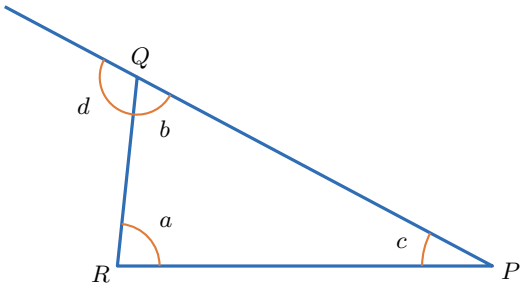
(B) 18

(C) 24

(D) 12

41. Medium · Desmos

In the figure shown, a, b and c are the three interior angles of a triangle, with $a = 3c$. Side PQ of the triangle is extended through Q, and the angle d that this extension makes with the side at Q measures 112°, so b and d together form a straight line. What is the value of $a + b$?



(A) 158

(B) 140

(C) 146

(D) 152

42. Medium · Desmos

What is the positive difference, in degrees, between the measure of an exterior angle of a regular octagon (8 sides) and the measure of an exterior angle of a regular decagon (10 sides)?

43. Medium · Desmos

At a certain time of day, a 6-meter flagpole casts a shadow that is 4 meters long. At the same time, a nearby building casts a shadow that is 30 meters long. What is the height, in meters, of the building?

44. Medium · No calculator needed

Triangle DEF is similar to triangle GHI, where D, E, and F correspond to G, H, and I, respectively. The measure of angle D is $48°$, and the measure of angle E is $63°$. What is the measure of angle H?

(A) $117°$

(B) $48°$

(C) $63°$

(D) $69°$

45. Medium · Desmos

In a triangle, the measures of the three interior angles are x, $2x$, and $(3x - 30)$ degrees. What is the value of x?

(A) 32

(B) 35

(C) 40

(D) 30

46. Medium · Desmos

Line ℓ is parallel to line m, and a transversal crosses both lines. One angle formed at line ℓ measures $(4x+5)°$, and the corresponding angle formed at line m measures $(6x-21)°$. What is the value of x?

(A) 15

(B) 18

(C) 11

(D) 13

47. Medium · Desmos

Two lines intersect, forming four angles. One angle measures $(5x+10)°$ and the angle vertical to it measures $(3x+30)°$. What is the value of x?

(A) 10

(B) 12

(C) 15

(D) 8

48. Medium · Desmos

A triangle has side lengths of 9, 14, and x, where x is an integer. What is the greatest possible value of x?

(A) 21

(B) 22

(C) 23

(D) 24

49. Medium · Desmos

Triangles ABC and DEF are similar. Side AB of triangle ABC corresponds to side DE of triangle DEF. If the length of $\overline{AB}$ is 12 units and the length of $\overline{DE}$ is 8 units, then the area of triangle DEF is equal to

(A) $\frac{16}{9}$ (area of triangle ABC)

(B) $\frac{2}{3}$ (area of triangle ABC)

(C) $\frac{4}{9}$ (area of triangle ABC)

(D) $\frac{9}{4}$ (area of triangle ABC)

50. Medium · Desmos

Two parallel lines are cut by a transversal. A pair of alternate interior angles measure $(6x - 8)°$ and $(4x + 12)°$. What is the measure, in degrees, of each of these angles?

51. Medium · Desmos

An exterior angle of a triangle measures $(8x-6)°$. The two remote interior angles measure $(3x+14)°$ and $(2x+19)°$. What is the measure, in degrees, of the exterior angle?

52. Medium · Desmos

In an isosceles triangle, the two base angles each measure $(x+20)°$ and the vertex angle measures $(4x-10)°$. What is the measure, in degrees, of the vertex angle?

53. Medium · Desmos

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ such that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD = 10$, $DB = 6$ and $AE = 15$, what is the length of $\overline{EC}$?

54. Medium · Desmos

Each interior angle of a regular polygon measures $140°$. How many sides does the polygon have?

55. Medium · Desmos

A trapezoid has parallel sides of length 10 and 24. A segment joins the midpoints of the two non-parallel sides. What is the length of this segment?

(A) 17

(B) 34

(C) 14

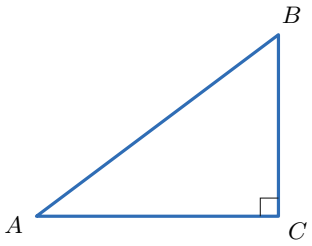
(D) 7

56. Medium · Desmos

In the xy-plane, a triangle has vertices $(0, 0)$, $(8, 0)$ and $(3, k)$, where $k > 0$. The area of the triangle is 20. What is the value of k?

57. Medium · Desmos

In right triangle ABC, the right angle is at C. If $\sin(A) = \frac{3}{5}$, what is the value of $\cos(B)$?



(A) $\frac{4}{5}$

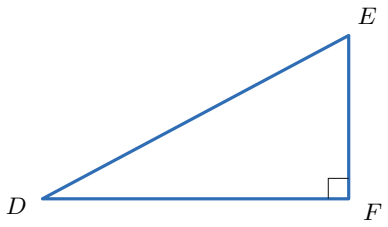
(B) $\frac{5}{3}$

(C) $\frac{3}{4}$

(D) $\frac{3}{5}$

58. Medium · Desmos

In right triangle DEF, the right angle is at F. If $\tan(D) = \frac{8}{15}$, what is the value of $\tan(E)$?



(A) $\frac{17}{15}$

(B) $\frac{8}{15}$

(C) $\frac{15}{8}$

(D) $\frac{8}{17}$

59. Medium · Desmos

If θ is an acute angle and $\sin(\theta) = \frac{5}{13}$, what is the value of $\tan(\theta)$?

(A) $\frac{12}{13}$

(B) $\frac{5}{12}$

(C) $\frac{12}{5}$

(D) $\frac{5}{13}$

60. Medium · No calculator needed

In a 30°-60°-90° right triangle, the side opposite the 30° angle has length 7. What is the length of the hypotenuse?

(A) 7

(B) $7\sqrt{2}$

(C) $7\sqrt{3}$

(D) 14

61. Medium · No calculator needed

What is the value of $\cos(60°) + \sin(30°)$?

(A) 1

(B) $\sqrt{3}$

(C) $\frac{1}{2}$

(D) $\frac{\sqrt{3}}{2}$